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Emotion rigidity in adolescents prospectively predicts future depressive symptoms assessed via self-report and clinical interview

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ABSTRACT

Adolescence is a critical developmental period marked by an elevated risk for Major Depressive Disorder (MDD). Identifying predictive risk factors is crucial for the development and timely delivery of preventative interventions. One potential risk factor is maladaptive and rigid emotion dynamics, which findings suggest is linked with depressive symptoms. Accordingly, to identify individuals at risk, the current study investigated whether emotional rigidity is a prospective predictor of depressive symptoms in adolescence using both self-reported and clinician-rated depression assessments. Adolescents (n = 117, aged 12–18) completed self-reported and clinician-rated measures of depressive symptoms at baseline and every 6 months for two years. Additionally, they received one month of ecological momentary assessment (EMA) surveys at baseline, reporting their emotions four times a day (n = 10,196). Multilevel vector autoregressive (mlVAR) models estimated temporal and contemporaneous density (operationalization of emotion rigidity) of subject-specific networks. Higher temporal and contemporaneous network density, as well as an aggregate density score, predicted greater future depressive symptoms controlling for baseline symptoms, as measured by either participants' self-report or clinician reports. Density emerged as a stronger predictor of future depression than any specific emotion inertia (i.e., predictability of one emotion state). Additionally, and supporting incremental predictive validity, temporal and aggregated density scores predicted future clinician-rated depression, and the aggregated score predicted self-reported depression, over and above mean levels of negative and positive affect. Collectively, these findings may have significant clinical implications for early identification and targeted preventative interventions among youth at risk of future depression by virtue of relatively rigid emotion dynamics.

Major Depressive Disorder (MDD), a severe emotional disorder, is now the leading cause of disability worldwide (Friedrich, 2017; World Health Organization, 2020). With an increase of about 30 % due to the COVID-19 pandemic, over 350 million people are currently suffering from depression (Santomauro et al., 2021). Adolescents are known to be at elevated risk to develop depression. During the adolescent years, depression rates rise dramatically, and by age 18, approximately 15 % of all teens will have developed MDD (Avenevoli et al., 2015). MDD is not only common, but it also bears extensive personal and social costs, including an elevated risk of suicide (Smith & De Torres, 2014). Given the significant prevalence and costs of MDD, identifying risk factors that increase vulnerability to depression and serve as early warning signs is crucial for developing effective and timely interventions. This is particularly important in research on adolescence, as this developmental period offers a window of opportunity to intervene prior to the first onset of depression. Notably, a prior depressive episode is one of the strongest predictors of future depression (Burcusa & Iacono, 2007).

One potential risk factor for the onset of depression is maladaptive and rigid emotion dynamics (Stange et al., 2017). Multiple emotion theories emphasize the importance of flexible emotion dynamics in processing information and adaptively reacting to the environment (Barrett et al., 2007; Frijda, 1988). Humans evolved to navigate a dynamic and unpredictable environment. To survive and thrive, our brains must continuously anticipate future demands and adapt our responses in real-time (Hutchinson & Barrett, 2019). In this ongoing process, emotions serve as crucial signals, indicating the nature of the challenges

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individuals encounter and how they can effectively manage (Bennett et al., 2022; Cunningham et al., 2013). Emotion rigidity, on the contrary, reflects inappropriate persistence in the same emotional state across time despite changes in circumstances. This can hinder one's ability to adjust to changing circumstances, potentially creating a cycle of functional impairment that may, over time, contribute to the development of MDD (Houben et al., 2015).

The view of emotion rigidity as antecedent of MDD corresponds with the *dynamic systems theory* perspective of the emotional system (van de Leemput et al., 2014). From this perspective, mental disorders such as depression are complex systems prone to sudden transitions between stable states (e.g., healthy vs. depressed). The fragility of the system develops gradually until a tipping point is reached, at which time the system abruptly shifts from one state (e.g., a healthy state) to another (e.g., a depressed state). These shifts are often preceded by early warning signals, such as "critical slowing down," which occurs when the system requires increased periods of time to recover from small perturbations (van de Leemput et al., 2014; Wichers, 2014). Emotion rigidity can be seen as an early warning signal, that could potentially predict depressive episodes before they occur. Specifically, a healthy system is able to quickly bounce back from a negative emotion (low rigidity), whereas a system nearing a tipping point towards depression may exhibit increased rigidity, struggling to return to a balanced state after even small emotional disturbances (Bringmann et al., 2022).

Evidence supporting the link between emotion rigidity and depression comes mostly from research on emotional inertia (for a review see Koval & Kuppens, 2024). Emotional inertia is defined as the extent to which specific emotions (or aggregated emotions, such as negative affect) persist over time (Kuppens et al., 2010). It is typically measured using autocorrelation, indicating how well an emotional state at one point in time predicts the same state at a later point. In simpler terms, it captures the extent to which someone is "stuck" in a particular emotion. Inertia of negative emotions has been shown to be related to the severity of depressive symptoms in non-depressed, healthy samples (Koval et al., 2012; 2013Kuppens et al., 2010) and to MDD diagnosis (Kuppens, Allen, et al., 2010; Nelson et al., 2020, see Thompson et al., 2021 for an exception). A meta-analysis by Houben et al. (2015) confirmed these findings, which were further replicated in children and adolescents (Reitsema et al., 2022, 2023). However, recent studies have shown that associations between inertia and measures of psychological functioning are not necessarily robust when statistically controlling for other more basic features of affect, such as its mean level and variability (Bos et al., 2019; Dejonckheere et al., 2019; Houben & Kuppens, 2020).

Most importantly, these previous studies have two major limitations. First, in most studies emotional rigidity was assessed during a depressive episode, making it impossible to determine whether emotional rigidity is a consequence, correlate, or a preceding risk factor for MDD (for exception see Kuppens et al., 2012). One possibility is that depressive symptoms slow down the way an adolescent's emotions unfold across time, making emotional rigidity a consequence of MDD (Nelson et al., 2020). Alternatively, emotional rigidity could reflect an early form of emotional disturbance or dysregulation that precedes and serves as a vulnerability to MDD (Kuppens & Verduyn, 2017). Prominent researchers have recently suggested that emotional inertia may be more characteristic of increased vulnerability to future depressive episodes rather than a feature of emotion dynamics reflecting current or past depression (Koval et al., 2021; Koval & Kuppens, 2024). However, further research is needed to explore these alternative explanations and determine the causal direction between emotional rigidity and MDD.

Second, studies exploring emotional rigidity through the lens of inertia are limited in their ability to capture the full complexity of the emotional system. This is because inertia primarily focuses on the predictability of a *single* emotion over time, neglecting the dynamic interplay of various emotions that characterizes the fullness of human experience. For example, specific negative emotions like sadness can reactively trigger various other emotions, such as disgust or anger. If this reactive pattern persists consistently, it could also indicate emotional rigidity. An alternative approach to measure emotional rigidity that overcomes this limitation is network analysis, an approach that simultaneously considers the dynamics across a range of emotions (Bar-Kalifa and Sened, 2020). When analyzing data collected repeatedly over time, this approach also enables researchers to construct networks to explore dynamics across different time scales. This includes revealing relationships between emotions at a single point in time (*contemporaneous networks*) and across consecutive time points (*temporal networks*) (Epskamp et al., 2018). Each network can be characterized by its density level, which indicates the overall strength of connections between the network's variables. A temporal network with strong connections between individuals' emotions over time (high temporal density) might signal an emotional system stuck in a repetitive loop, unable to respond dynamically to environmental changes (Pe et al., 2015). A denser contemporaneous network, characterized by strong associations between individual emotions, could suggest a less granular, less nuanced emotional system, potentially indicative of a maladaptive pattern. That is, high temporal network density indicates that the person is "stuck" in the same pattern of emotions over time, while high contemporaneous network density indicates that the person's emotions are "sticky," with the same emotions tending to appear together at the same time. A recent study investigating depressive symptoms demonstrated that these two types of network structure revealed distinct and valuable information about the nature of emotional dynamics (Bose et al., 2017). Critically, previous studies that used this approach to explore the association between emotion rigidity and depression found that temporal emotion network density was related to depression diagnosis in a sample of adults (Pe et al., 2015; Shin et al., 2022) and to levels of depression in a sample of adolescents (Lydon-Staley et al., 2019). However, these studies examined depression at the same time rather than investigating whether emotion rigidity serves as an early warning signal that can predict future depression in currently healthy individuals.

The current study

The current study aims to investigate whether emotion rigidity predicts future depressive symptoms in adolescents when controlling for baseline depressive symptoms. Exploring emotional dynamics is crucial within this population, as this developmental period is characterized by both increased difficulties in emotion regulation and the acquisition and strengthening of regulatory abilities (Hollenstein & Lanteigne, 2018). Additionally, depression rates surge during adolescence, making it the most common period for the onset of depression relative to childhood or adulthood (Avenevoli et al., 2015). This creates a critical period during which recognizing maladaptive emotion dynamics and targeting them before the onset of depression could inform the development and delivery of prevention efforts.

To address the abovementioned limitations of prior research, the current study uses network analysis to measure emotion rigidity, while also tracking changes in depressive symptoms over a two-year period following the assessment of rigidity. The study of emotion dynamics requires repeated assessments of the individual across contexts, circumstances, and/or time, so as to provide ample opportunity to capture changing responses (Silk et al., 2011). Accordingly, we used smartphone-delivered ecological momentary assessment (EMA) to repeatedly sample emotions (via brief smartphone surveys) as they occurred in the daily life of adolescents (Hormuth, 1986). In addition, we tested our hypotheses using both self-report and clinician-rated measures of depression as our dependent variable.

1. Method

1.1. Participants

The initial sample included 151 adolescents aged 12–18 (88 female,

63 male; $m_{age} = 14.2$, $SD = 1.88$) recruited from the greater Boston area. Participants came from two similar studies conducted in the same lab ($N_{study1} = 37$, $N_{study2=}114$) investigating adolescents' risk for developing depression, either by virtue of elevated anhedonia or a biological parent with a history of depression. Across both studies 90 participants were healthy control (who were not at elevated risk to develop depression), 61 were at elevated risk to develop depression. Both studies had the following general exclusion criteria: past or present diagnosis of major depressive disorder, bipolar disorder, psychotic disorder, neurodevelopmental disorder, or hypothyroidism, present diagnosis of moderate or severe alcohol or substance use disorder, past diagnosis of severe alcohol or substance use disorder, anorexia nervosa, or bulimia nervosa (confirmed by a clinical interview for DSM-5 disorders). Participants at risk for developing depression either had a biological parent meeting DSM-5 criteria for MDD (current or past), or elevated levels of anhedonia, as defined by a SHAPS (Snaith-Hamilton Pleasure Scale) score $\geq$3 (based on the original [binary] scoring) and an elevated anhedonia score (>1) on the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS; Kaufman et al., 1997). Healthy control participants had no first-degree relatives with past or present MDD, bipolar disorder, or a psychotic disorder. In the anhedonia-related study, healthy controls had minimal anhedonia, i.e., all had a 0 on the SHAPS and a 1 on the K-SADS. Participants on psychotropic medication were excluded from the study, with the exception of anhedonic participants who were on a stable dose of selective serotonin reuptake inhibitors (SSRI) for at least a month. None of the participants in our final sample were on medication. Participants were part of a larger study that included an fMRI component, so common fMRI contraindications were also exclusionary (see Webb et al., 2022 for more information). Demographic and baseline characteristics of the sample are shown in Table 1.

1.2. Procedure

The study was approved by the Mass General Brigham (MGB) Institutional Review Board (IRB), protocol numbers: 2018P001486 and 2015P001303. Participants initially completed a **screening session with their parents**, during which parents completed **symptom questionnaires** and participated in a **structured clinical interview, including the Mini-International Neuropsychiatric Interview (MINI) and the Family History Screen**, to assess family history and the presence of psychopathology. Following this screening, participants completed a **baseline session**, which was conducted either **in-person or over Zoom**. At this session, adolescent participants were administered a semi-structured clinical interview, the K-SADS. Afterwards, they completed a set of online questionnaires which included the Center for Epidemiological Studies Depression Scale (CES-D; Radloff, 1977) and SHAPS. Participants then installed the MetricWire app on their smartphones in order to complete 30 days of Ecological Momentary Assessment (EMA). In the first sample, EMA prompts were randomly scheduled within predefined time windows to capture affect fluctuations throughout the day: Morning (7:00–8:00 a.m., before school), Midday (11:30 a.m.–12:30 p.m., lunch break), Afternoon (3:00–6:00 p.m., after school), and Evening (6:00–9:00 p.m., before bedtime). In the second sample, EMA prompts were randomly administered four times per day between 6:00 a.m. and 8:00 p.m., with a minimum interval of 1 h between prompts to ensure adequate spacing. Each EMA survey took approximately 1–2 min to complete and included questions about affect, sleep, stress, cognitions, and recent activities. Participants completed the surveys on their personal smartphones using the MetricWire app, which securely encrypted and uploaded responses to a HIPAA-compliant server. At 6-, 12-, 18-, and 24-months after the baseline session, participants completed the baseline self-report measures and K-SADS (via Zoom).

Table 1
Demographic and clinical characteristics of the sample.

Sample Characteristics	N	%
Biological Sex		
Female	88	55.3
Male	63	41.7
Race		
American Indian or Alaska Native	1	0.7
Asian	12	8.0
Black or African American	8	5.3
Native Hawaiian or Other Pacific Islander	1	0.7
White	118	78.2
Other	1	0.7
More than one race	13	8.6
Ethnicity		
Hispanic or Latino	11	7.3
Not Hispanic or Latino	140	92.7
Current Diagnoses (DSM-V)		
Generalized Anxiety Disorder	5	3.3
Social Anxiety Disorder	2	1.3
Panic Disorder	1	0.7
Specific Phobia	0	0.0
Attention-Deficit/Hyperactivity Disorder	1	0.7
Oppositional Defiant Disorder	0	0.0
Obsessive Compulsive Disorder	1	0.7
	M	**SD**
Age (in years)	14.2	1.88
Family Income (dollars)	163,673	125,636.38

2. Measures

2.1. EMA

Six adjectives (three positive and three negative) were chosen from the Positive and Negative Affect Scale for Children (PANAS-C; Laurent et al., 1999). Participants rated their current positive affect (PA) and negative affect (NA) indicating the extent to which they were feeling "happy", "interested", and "excited" (for PA) and "sad", "angry", and "nervous" (for NA) immediately before receiving the survey. While the first sample used a 0–100 sliding scale, the second sample used a 0–4 scale. To ensure consistency, we multiplied the scores of the second sample by 25 before analysis. This transformation did not affect the results.

2.2. Baseline and follow-up assessments

2.2.1. Center for Epidemiological Studies Depression Scale (CES-D)

Participants completed the CES-D, a 20-item self-report scale that measures depression symptom severity at baseline and at each follow-up (Radloff, 1977). The scale assesses how often participants experience a variety of depressive symptoms in the past week on a 4-point Likert scale ranging from 1, "Rarely or none of the time (less than 1 day)" to 4, "Most or all of the time (5–7 days)." The internal consistency was assessed using Cronbach's alpha, which yielded a value of $\alpha = .92$.

2.2.2. Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS)

Participants were administered the K-SADS, a semi-structured clinical interview that assesses current and lifetime history of psychiatric disorders according to the Diagnostic and Statistical Manual of Mental Disorders–5 (DSM–5) criteria at baseline and at each follow-up (Kaufman et al., 1997). The interview was conducted under the supervision of a licensed clinical psychologist (CAW) by Bachelor's-level research assistants or a postdoctoral fellow either over Zoom or in person. Clinician-rated future depression scores from the K-SADS were calculated based on the two core symptoms of depression, depressed mood and anhedonia, since, per the measure's standardized administration, additional symptom questions were only administered to participants who scored above a certain threshold on these items. Interrater reliability for the K-SADS was not assessed in this study, however, parallel study conducted at the same time with the same interviewers found

excellent reliability. In that study, a random subset of 25 % of the K-SADS interviews was independently rated by a PhD-level Clinical Psychologist, yielding perfect interrater reliability (κ = 1.0) (Webb et al., 2024).

2.3. Statistical analysis

The median number of EMA observations was 94 (Mean = 89.32, Min = 33, Max = 122). The Welch Two-Sample *t*-test indicated no significant difference in the mean number of EMA observations between groups, t(120.49) = −0.079, p = .937, 95 % CI [−9.02, 8.33], with highly similar means ($M_{high\text{-}risk}$ = 89.52, $M_{low\text{-}risk}$ = 89.18), and the Wilcoxon rank-sum test further confirmed that the median number of EMA observations did not significantly differ between groups, W = 2652.5, p = .727. Data preprocessing was performed, in this step participants with extremely low variance (less than 5 % unique values for each emotion variable across time points) were excluded. In addition, to ensure a more robust network, we included only participants who completed over 30 EMA assessments (Burger et al., 2022). Two participants were excluded due to having completed fewer than 30 assessments, and the remainder were excluded based on low variance. This resulted in the inclusion of 117 participants who completed 10,196 EMA assessments in subsequent analyses.

To estimate person-specific temporal and contemporaneous networks, we utilized a two-step multilevel vector autoregression (mlVAR; Epskamp et al., 2018). This model extends the vector autoregression (VAR) by allowing the estimation of individualized emotional networks within a group of subjects. In the mlVAR model, all variables at time point (t) are regressed on variables from the previous assessment (t-1). To separate within- and between-person variance, predictors are centered around each person's mean, with sample means added as predictors. Residuals from this step are used in the second step to explore contemporaneous relationships. This procedure results in two types of networks. The *temporal network* captures the average lagged within-person relationships between emotions measured at successive assessments, controlling for other lagged associations. The *contemporaneous network* represents the within-person relationships between the emotions at the same assessment timepoint, controlling for each other and the temporal effects. To constrain estimates to *within*-day effects, the lag from the last measurement of one day to the first measurement of the subsequent day were excluded from analysis. This is particularly relevant as measuring changes within the same day may capture short-term fluctuations, which can differ from changes observed across consecutive days, potentially influencing the interpretation of results (Shin et al., 2022).

After estimation of the dynamic network models, network density was calculated by summing the absolute values of each subject's network effects. This yielded two types of density: temporal density and contemporaneous density. In addition, to capture the overarching construct of emotional system rigidity, we also summed the temporal and contemporaneous rigidity scores ("aggregate density"), expecting that higher levels in both types would indicate greater emotional rigidity and may be a relatively robust predictor of future depression risk (in comparison to temporal or contemporaneous rigidity scores on their own).

First, we tested the association between the three density parameters and baseline depressive symptoms measured using self-report (CES-D) and clinician ratings (K-SADS). Second, we tested whether the three density parameters predicted future levels of depression at follow-up assessments. To create a variable representing self-reported future depression, we selected the highest depression score for each individual across all available follow-up assessments. Fig. S1 in the supplement shows the distribution of time points included in this new maximum depression variable. To create a variable for clinician-rated future depression, we calculated the sum of the scores from the two criterial symptoms of depression (i.e., depressed mood and anhedonia) over the follow-up assessments for each individual. This approach reflects the differences in measurement between the two scales: the CES-D provides a continuous measure of symptom severity, allowing for greater variability, while the K-SADS is a clinician-administered diagnostic tool that categorically assesses symptom presence, leading to reduced variability. Therefore, we used the maximum CES-D score to capture peak symptom severity and the sum of K-SADS scores over time to reflect cumulative depressive burden.

All analyses controlled for sex and risk group (i.e., high risk = having either elevated anhedonia or a parental history of depression; low risk = neither risk factor). When predicting future depression, we also controlled for baseline depression. Listwise deletion was used to manage missing data resulting in 112 and 101 participants included in the baseline and future prediction, respectively.

Only 4 participants had missing data at baseline, one for the K-SADS and three for the CES-D, therefore, we decided not to impute the data and use listwise deletion instead.

2.4. Transparency and openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. Data analysis was conducted in the R environment. The analytic code and additional materials are available online (https://osf.io/dm4w5/). Mclean Hospital and Mass General Brigham (MGB) require an IRB-approved signed data sharing agreement with the individual requesting the data. This study and its analyses were not preregistered; however, we closely adhered to the analytic approach and decision framework outlined by Kelley et al. (2023). This approach allowed for continuity and comparability with prior research.

3. Results

The mean, standard deviations, and correlation coefficients for the variables are presented in Table 2. Fig. 1 depicts the temporal network (a) and the contemporaneous network (b) of the full sample as well as examples of person-specific emotional networks. Panels c and d depict temporal networks, with one illustrating high density and another showcasing low density. Panels e and f show contemporaneous networks, again with examples of participants with high and low density.[1]

3.1. Relation between baseline depressive symptoms and emotion rigidity

Self-report. Baseline depressive symptoms, as measured with the CESD, were significantly associated with temporal network density (β = 0.20, *t* = 2.06, *p* = .042), contemporaneous network density (β = 0.25, *t* = 2.49, *p* = .014), and the aggregated density score (β = 0.28, *t* = 2.90 *p* = .005).

Clinical interview. Depressive symptoms, as measured using the K-SADS, was not significantly associated with any of the density scores, including the temporal network density (β = 0.10, *t* = 0.53, *p* = .60), contemporaneous network density (β = 0.05, *t* = 0.83, *p* = .41), or the aggregated density score (β = 0.08, *t* = 0.85, *p* = .40).

[1] As noted above, the sample was derived from two studies that were mostly similar, with the only difference being the criteria for the risk groups (see Participants section). No significant differences in either type of rigidity were found between the two samples (Temporal density: t = −0.11522, df = 43.235, p-value = 0.91; contemporaneous density: t = 2.0238, df = 39.547, p-value = 0.05). Additionally, including a variable representing the study in the analysis as a covariate or in interaction with the density measures did not change any of the results.

Table 2
Emotion network density and symptoms of depression scores: Correlations and descriptive statistics.

	1	2	3	4	5	6	7
Temporal density	1						
contemporaneous density	0.26**	1					
Aggregated density	0.82***	0.76***	1				
Baseline CESD depression	0.27**	0.24**	0.33***	1			
Follow-up CESD depression	0.39***	0.31**	0.45***	0.65***	1		
Baseline K-SADS depression	0.14	0.10	0.15	0.60***	0.45***	1	
Follow-up K-SADS depression	0.38***	0.21*	0.38***	0.61***	0.63***	0.65***	1
Mean (SD)	2.95 (0.73)	4.12 (0.64)	7.07 (1.09)	9.33 (8.66)	16.4 (11.1)	1.55 (1.11)	2.48 (2.67)

3.2. Emotion rigidity predicting future depression

Self-reported depression. In the following analysis, in addition to controlling for age and risk group, we controlled for baseline severity of depressive symptoms. Results are presented in Fig. 2 and Table 3. Greater temporal ($\beta = 0.21$, $t = 2.57$, $p = .012$) and contemporaneous ($\beta = 0.16$, $t = 2.05$ $p = .043$) network density, as well as the aggregated score ($\beta = 0.24$, $t = 3.04$, $p = .003$), significantly predicted higher future depression scores. When running a model that included only contemporaneous, temporal, or aggregated density as a predictor of future depression, these variables accounted for 9 %, 15 %, and 19 % of the variance in the future depression score, respectively.

We repeated this analysis using only the depression (CESD) symptom scores at the one-year follow-up as the dependent variable (rather than the max depression score across all available follow-up timepoints, which may be influenced by the number of available follow-up timepoints for each individual). We chose this timepoint because it represents the furthest assessment at which we had a sufficient number (n = 71 for self-report, n = 74 for clinical ratings) of participants with observations (1-year depression scores) to conduct the analysis. Details on the sample size at each follow-up time point are available in Table S1 of the supplement. Results revealed the same pattern, with both temporal and contemporaneous network density, as well as the combined score, significantly predicting higher one-year follow-up depressive symptoms (temporal: $\beta = 0.23$, $t = 2.46$ $p = .016$; contemporaneous: $\beta = 0.26$, $t = 2.92$ $p = .005$; aggregated score: $\beta = 0.31$, $t = 3.44$ $p = .001$).

3.3. Clinician-rated depression

Temporal network density ($\beta = 0.24$, $t = 3.20$, $p = .002$) and the aggregated density scores ($\beta = 0.25$, $t = 3.30$ $p = .001$) significantly predicted higher future criterial depressive symptoms as determined by the K-SADS clinical interview. Contemporaneous network density did not significantly predict future criterial depressive symptoms ($\beta = 0.14$, $t = 1.87$, $p = .064$). In addition, whereas temporal network density and the aggregated scores predicted K-SADS one-year follow-up criterial depressive symptoms (n = 74, temporal: $\beta = 0.27$, $t = 2.84$, $p = .006$; aggregated score: $\beta = 0.24$, $t = 2.60$ $p = .011$), the contemporaneous density score did not predict criterial depressive symptoms ($\beta = 0.09$, $t = 1.04$, $p = .30$). See supplement (Table S2) for additional analyses using alternative K-SADS depression scoring, which yielded a very similar pattern of findings.

Sensitivity analysis. Given that both temporal and contemporaneous network density each predicted future depression (though contemporaneous density only marginally predicted clinical rating depression measure score), we proceeded to simultaneously include both in the same model to ascertain whether one or both explained a unique portion of the variance. Analysis showed that only the temporal network density remained significant across outcome measures (Table S3 in the supplement). Next, since we treated the two depression measures differently (i.e., using the maximum CES-D score regardless of timing and the total K-SADS scores over time), we tested whether density would predict these measures using a consistent approach. Using maximum scores for the K-SADS would remove important variability, so we instead decided to sum the follow-up scores for both measures. Under this approach, we found that both temporal density and the aggregated scores (but not contemporaneous density) predicted the summed CES-D scores (see Supplement Table S4).

We also examined whether the density scores predicted future depressive symptoms above and beyond the mean levels of positive affect and negative affect. When controlling for mean positive and negative affect, the aggregated density score significantly predicted self-reported depression symptoms (CESD). In addition, both temporal network density and the aggregated score significantly predicted depression assessed through clinical interview (K-SADS) (see Table 4). Notably, when predicting depression at one-year follow-up, the pattern remained consistent for most measures. However, contemporaneous density emerged as a significant predictor of self-reported depressive symptoms (see online Supplementary Table S5). Finally, we repeated the analysis controlling for the dispersion (SD) of the emotion scores. All density metrics remained significant predictors of both self-report and clinician-rated depression measures (see online Supplementary Table S6).

Comparing density scores with inertia of individual emotions. We calculated the inertia (autoregressive) score for each emotion item separately. The correlation between inertia in each emotion, depression measures, and density scores is depicted in Fig. 3. All emotion inertia scores were significantly correlated with temporal density. However, depression scores (either baseline or future self-reported or clinical interview scores) were not associated with inertia scores with the exception inertia in sadness being significantly correlated with future clinical interview (K-SADS) depressive scores. We reran the analyses with either the temporal, contemporaneous, or aggregated density score as a predictor of follow-up depressive scores, while controlling for sadness inertia score. Network density scores remained significant predictors of future depression in all models (see Online Supplementary Table S7). However, sadness inertia only predicted future depression in the model including contemporaneous density as a predictor.

Testing Robustness and Specificity with Anxiety. To assess the robustness of our findings, we conducted additional analyses controlling for baseline Multidimensional Anxiety Scale for Children (MASC) scores. The pattern of results remained unchanged, reinforcing the stability of our findings (see Table S8 in the supplement). To further examine the specificity of our findings, we tested whether emotional rigidity predicted future anxiety symptoms. As shown in Table S9, none of the rigidity indices significantly predicted future anxiety, suggesting that this mechanism may be more specifically associated with depressive symptomatology. To account for the zero inflation in the K-SADS (55.56 % zeros), we conducted a sensitivity analysis comparing multiple models and identified the Hurdle **Negative Binomial** model as the best fit. This model showed that temporal (but not contemporaneous) density significantly predicts higher K-SADS scores among those with depressive symptoms but does not predict symptom presence. Full description of the analyses and results are presented in Table S10 and S11. In addition, we re-ran the analyses using a less restrictive threshold for excluding participants with low emotion variability. This resulted in no participants being removed due to low variability, the overall pattern of results remained consistent (see Table S12 in the supplement). Finally, details

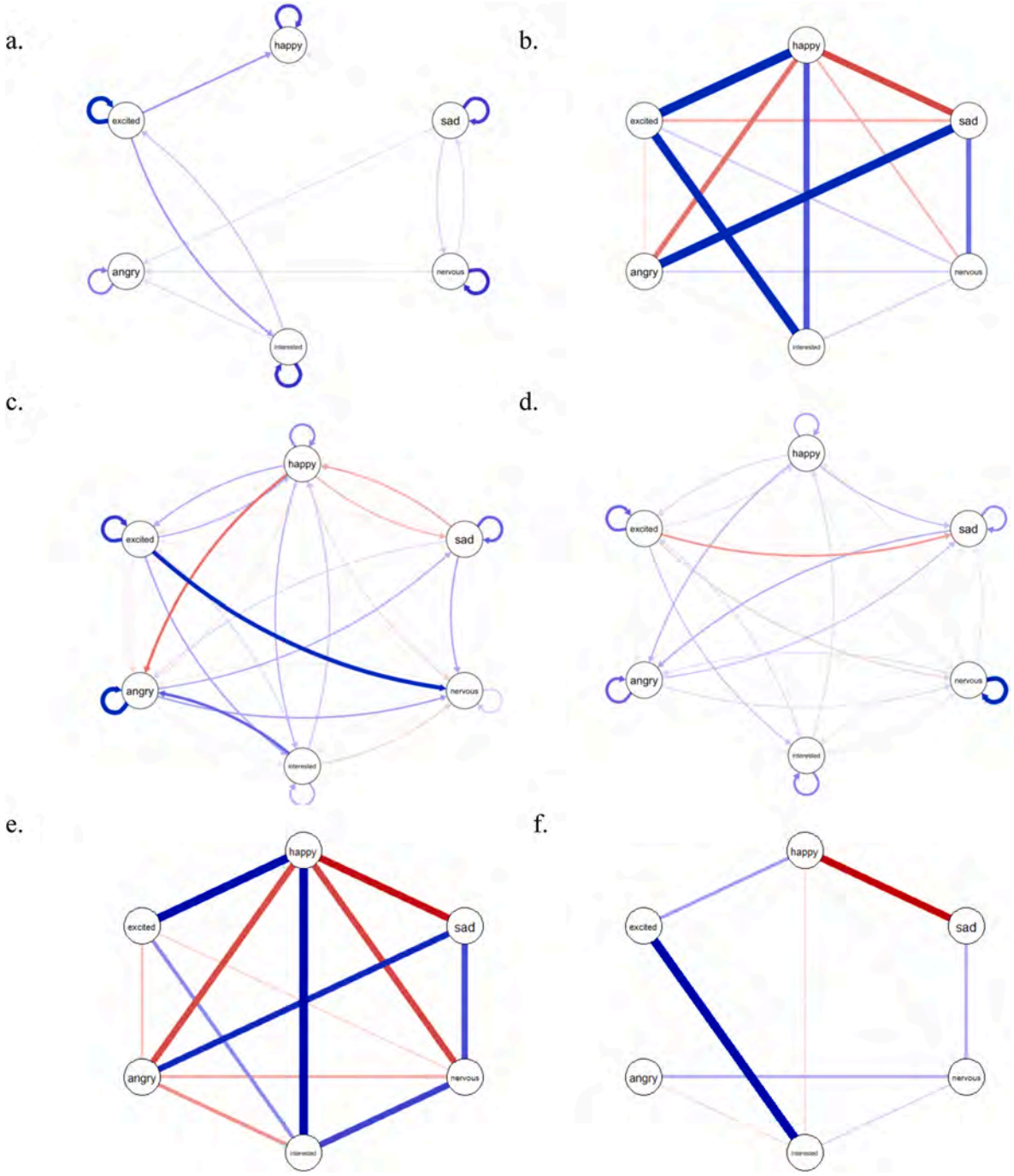


Fig. 1. Visualization of the group-level emotion network and person-specific emotional networks (high vs. low Density)
Note. (a) and contemporaneous (b) relations among 6 emotions assessed four times a day over the course of a month. These relations were estimated with a multilevel vector-autoregressive model and depicted as a graph, with nodes representing variables and edges (lines) representing partial correlations among variables. Thicker and more saturated edges depict stronger relations. In temporal network, high density indicates strong interconnectedness between emotions across time points. In contemporaneous density high density indicates strong co-occurrence of emotions at a single time point.
Panel a: group level temporal density. Panel b: group level contemporaneous density. Panel c: example of an individual temporal networks with high density. Panel d: example of an individual temporal networks with low density. Panel e: example of an individual contemporaneous networks with high density. Panel f: example of an individual contemporaneous networks with low density.

on the relationship between risk group, rigidity, and depressive symptom trajectories, including additional analyses, are provided in the supplement.

4. Discussion

This study is the first to employ network analysis to investigate whether emotional rigidity predicts future depression in healthy adolescents. Identifying emotional processes contributing to the onset of depression is crucial, particularly in this age group, as early depression onset often predicts a more chronic trajectory later in life (Zisook et al., 2007). This knowledge could be used to design preventive interventions targeting maladaptive emotion dynamics which may ultimately reduce the risk of depression onset in youth.

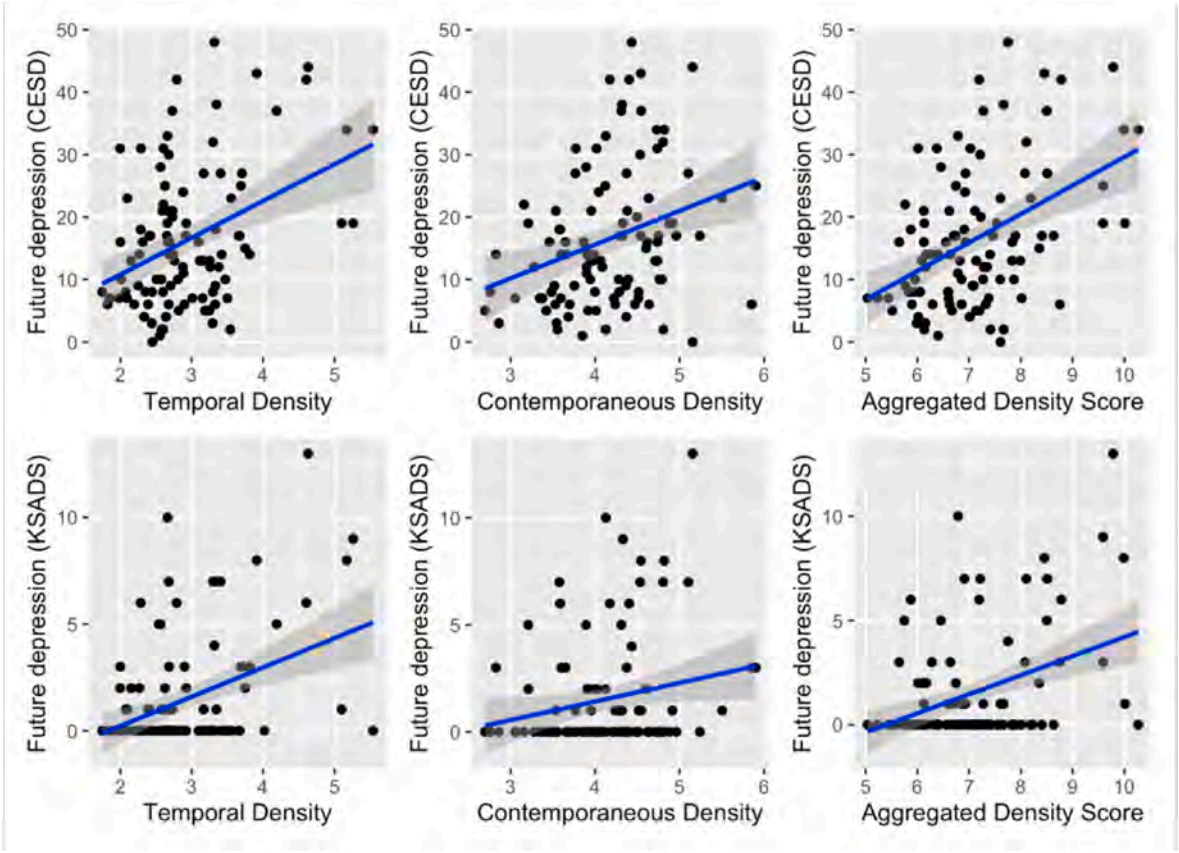


Fig. 2. Regression results of temporal, contemporaneous and aggregated density scores predicting depressive symptoms at follow-up.

Table 3
Regression Analyses results, emotional network density predicting depressive symptoms at follow up.

Outcome	Max future self-reported depression (CESD)					Future clinician-rated depression (K-SADS)				
Predictors	Est.	Std.	CI	t	p	Est.	Std.	CI	t	p
Temporal density										
Intercept	−0.00	0.07	−0.15–0.15	0.08	0.937	−0.00	0.07	−0.14–0.14	−2.68	**.009**
Baseline depression	0.56	0.08	0.39–0.73	6.64	**<.001**	0.55	0.08	0.40–0.70	7.17	**<.001**
Gender	−0.06	0.08	−0.22–0.10	−0.76	0.448	−0.07	0.08	−0.22–0.09	−0.86	0.395
Group	0.03	0.08	−0.14–0.19	0.32	0.752	0.11	0.08	−0.05–0.27	1.40	0.165
Temporal density	0.21	0.08	0.05–0.37	2.57	**.012**	0.24	0.08	0.09–0.39	3.20	**.002**
	N = 101, R^2/R^2 adjusted 0.469/0.446					N = 101, R^2/R^2 adjusted 0.510/0.489				
Contemporaneous density										
Intercept	−0.00	0.07	−0.15–0.15	−0.50	0.620	−0.00	0.07	−0.15–0.15	−1.63	0.107
Baseline depression	0.56	0.09	0.39–0.73	6.45	**<.001**	0.56	0.08	0.40–0.72	7.08	**<.001**
Gender	−0.07	0.08	−0.23–0.09	−0.89	0.377	−0.08	0.08	−0.24–0.07	−1.07	0.286
Group	0.08	0.08	−0.08–0.25	0.99	0.325	0.16	0.08	0.00–0.32	2.01	**.048**
Contemporaneous density	0.16	0.08	0.00–0.31	2.05	**.043**	0.14	0.07	−0.01–0.29	1.87	0.064
	N = 101, R^2/R^2 adjusted 0.456/0.433					N = 101, R^2/R^2 adjusted 0.48/0.45				
Aggregated density										
Intercept	−0.00	0.07	−0.15–0.15	−1.43	0.155	−0.00	0.07	−0.14–0.14	−3.01	**.003**
Baseline depression	0.53	0.08	0.36–0.70	6.26	**<.001**	0.55	0.08	0.40–0.70	7.17	**<.001**
Gender	−0.06	0.08	−0.21–0.10	−0.75	0.455	−0.07	0.08	−0.22–0.08	−0.89	0.375
Group	0.04	0.08	−0.12–0.21	0.55	0.582	0.12	0.08	−0.04–0.27	1.52	0.131
Aggregated density	0.24	0.08	0.08–0.40	3.04	**.003**	0.25	0.07	0.10–0.39	3.30	**.001**
	N = 101, R^2/R^2 adjusted 0.482/0.460					N = 101, R^2/R^2 adjusted 0.513/0.492				

Note. Standardized coefficients (β) are reported.

4.1. Concurrent depression

In our concurrent models, findings differed based on the depression outcome measure. Higher self-reported depressive (CESD) symptoms were significantly correlated with greater temporal and contemporaneous density as well as greater aggregated density score. In contrast, the clinician-rated measure of depression (K-SADS) was not correlated with any of the density scores. The finding of a significant correlation between contemporaneous density and self-report depressive symptoms corresponds with findings from research on emotion differentiation, which is defined as the ability to make nuanced distinctions between same-valenced emotions (Barrett et al., 2001). In essence, by assessing how strongly one emotion predicts another, contemporaneous density, similar to emotion differentiation, measures the ability to distinguish

Table 4
Regression Analyses results, emotional network density predicting depressive symptoms at follow up controlling for mean positive and negative affect.

Outcome	Max future self-reported depression (CESD)					Future clinician-rated depression (K-SADS)				
Predictors	Est.	Std.	CI	t	p	Est.	Std.	CI	t	p
Temporal density										
Intercept	−0.00	0.07	−0.14–0.14	1.07	0.289	−0.00	0.07	−0.14–0.14	−0.68	0.495
Baseline depression	0.49	0.08	0.33–0.66	5.96	**<.001**	0.53	0.07	0.38–0.67	7.20	**<.001**
Mean PA	−0.11	0.08	−0.26–0.04	−1.49	0.140	−0.21	0.07	−0.35–−0.07	−2.94	**.004**
Mean NA	0.21	0.08	0.04–0.38	2.52	**.013**	0.12	0.08	−0.04–0.28	1.49	0.139
Temporal density	0.14	0.08	−0.02–0.30	1.71	0.091	0.23	0.08	0.07–0.38	2.84	**.005**
	N = 101, R^2/R^2 adjusted 0.508/0.487					N = 101, R^2/R^2 adjusted 0.547/0.528				
Contemporaneous density										
Intercept	−0.00	0.07	−0.14–0.14	0.51	0.609	0.00	0.07	−0.14–0.14	0.13	0.897
Baseline depression	0.51	0.08	0.34–0.67	6.12	**<.001**	0.55	0.08	0.40–0.70	7.29	**<.001**
Mean PA	−0.10	0.08	−0.25–0.06	−1.27	0.207	−0.20	0.07	−0.35–−0.05	−2.62	**.010**
Mean NA	0.25	0.08	0.09–0.41	3.05	**.003**	0.20	0.08	0.05–0.36	2.67	**.009**
Contemporaneous density	0.09	0.08	−0.07–0.24	1.09	0.277	0.06	0.08	−0.09–0.21	0.83	0.408
	N = 101, R^2/R^2 adjusted 0.499/0.478					N = 101, R^2/R^2 adjusted 0.48/0.45				
Aggregated density										
Intercept	−0.00	0.07	−0.14–0.14	−0.22	0.155	−0.00	0.07	−0.14–0.14	−1.33	0.187
Baseline depression	0.49	0.08	0.32–0.65	5.91	**<.001**	0.54	0.07	0.39–0.68	7.29	**<.001**
Mean PA	−0.10	0.08	−0.25–0.05	−1.33	0.455	−0.19	0.07	−0.34–−0.05	−2.68	**.009**
Mean NA	0.20	0.09	0.03–0.37	2.32	0.582	0.12	0.08	−0.04–0.28	1.49	0.140
Aggregated density	0.16	0.08	−0.01–0.33	1.92	**.003**	0.20	0.08	0.04–0.37	2.49	**.015**
		N = 101, R^2/R^2 adjusted 0.482/0.460					N = 101, R^2/R^2 adjusted 0.538/0.519			

Note. Standardized coefficients (β) are reported. NA: Negative Affect, PA: Positive Affect.

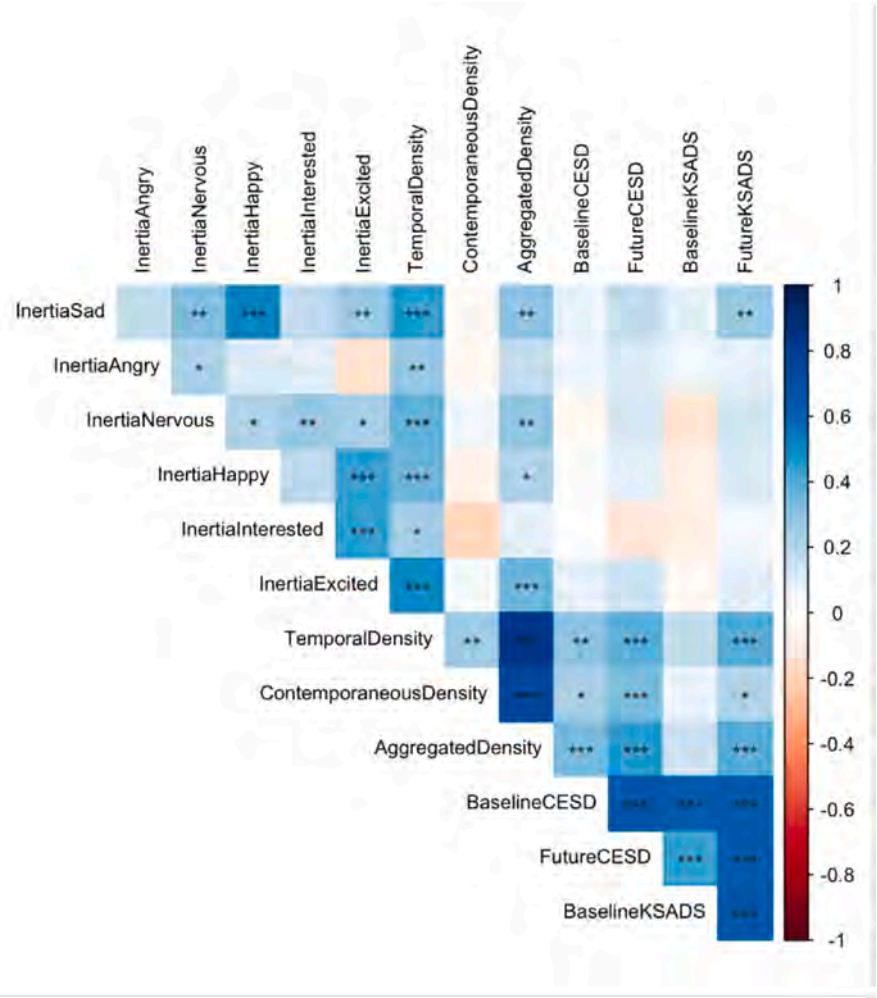


Fig. 3. The correlation between mean negative and positive affect, inertia in each emotion, depression measures, and density scores.

between different emotions experienced at the same time. Consistent with our results, a recent review confirms that low emotion differentiation is related to greater depressive symptoms in adolescents (Dunning, Wright, Bennett, Knight, & Dalgleish, 2022). The temporal density finding is consistent with a recent study in which adolescents completed daily diaries over a 21-day period, revealing a correlation between

temporal density and current depression levels (Lydon-Staley et al., 2019). The non-significant association between contemporaneous emotional rigidity measures and clinician-rated depression may be due to the limited variability observed in the latter measure at baseline (range 1–5, Mean = 1.55, SD = 1.11 at baseline vs. range 1–14, M = 2.48, SD = 2.67 at follow-up). Alternatively, self-report measures (relative to clinical interview measures) may be more sensitive to current internal emotional states because they directly capture the individual's subjective experience. Overall, the results provide some support for the concurrent relationship between emotion rigidity and depressive symptoms.

4.2. Future depression

In our prospective models, the association between emotion density and depressive symptoms measured at follow-ups were more robust. All density metrics (temporal, contemporaneous, and the aggregated scores) significantly predicted future depression assessed via self-report. In addition, both temporal and the aggregated scores significantly predicted future depression assessed via clinical interview ratings. These differences in predictive strength may, in part, result from differences in the reliability of the measures, as higher reliability reduces measurement error and enhances the ability to detect true associations. Additionally, differences in measurement approach between the CES-D, a dimensional self-report measure, and the K-SADS, a clinician-administered diagnostic interview, may have contributed to the observed pattern of results. The K-SADS is designed for diagnostic precision but may be less sensitive to capturing fluctuations in symptom severity. Furthermore, both temporal density and the aggregated score remained significant predictors of clinical interview ratings of depression after controlling for mean negative affect and positive affect. The aggregated score also predicted future self-report depression measure over and above the mean scores. These results support the notion that emotional rigidity reflects dynamic processes that increase vulnerability to depression, potentially serving as an early warning sign of depression onset risk (Koval & Kuppens, 2024).

Notably, density metrics did not predict future anxiety symptoms, which may suggest that emotional rigidity constitutes a disorder-specific vulnerability, particularly relevant to depression. This aligns with findings from Gilbert et al. (2020), who showed that a latent inflexibility construct (comprising perseverative thinking, personality rigidity, and emotional inertia) was more strongly associated with depression than with anxious arousal. Interestingly, emotional inertia was linked to anxiety in their study, possibly reflecting the broader role of sustained negative affect. Further research is needed to clarify whether rigidity reflects a depression-specific risk or a transdiagnostic process with varying expression across disorders.

The use of density in the context of network analysis allowed us to address limitations of previous studies that relied on inertia as a measure of emotional rigidity, thus providing a more comprehensive understanding of the dynamics of the emotional system. We demonstrated that only inertia in sadness was related to future depressive symptoms. Nevertheless, network density predicted future depressive symptoms above and beyond inertia. This highlights the importance of examining interactions within the emotional system and consider how emotions influence each other. Furthermore, network analysis allowed us to delineate a rigid emotional system with two dimensions. First, a rigid system can manifest as strong connections *between* emotions (e.g., when sadness arises, so too does anger and anxiety), reflecting poor emotion regulation. Second, a rigid system may get stuck in emotional states and take longer to recover. In the current study we evaluated rigidity at both contemporaneous (current) and temporal (over time) levels, representing these two possible dynamics of a rigid system. Our findings indicate that both temporal and contemporaneous maladaptive emotional patterns may contribute to depression development, but only temporal rigidity remains a significant predictor when analyzed together, suggesting it might be a stronger marker of future depression. To capture the overarching construct of emotional system rigidity, we constructed a latent variable by summing the temporal and contemporaneous rigidity scores. This decision was supported by two key observations. First, the moderate association between contemporaneous and temporal rigidity scores suggests they capture overlapping, yet distinct, aspects of this underlying construct. Second, the high correlation of the aggregated score with each individual measure strengthens the argument that it reflects a latent factor that captures the shared underlying construct. This aggregated rigidity score emerged as the most consistent indicator of future depression significantly predicting depression across all models. These findings suggest that a maladaptive and rigid emotional system may contribute to the risk of depression, potentially by hindering an individual's ability to effectively adapt to various situations. Consequently, interventions designed to reduce both temporal and contemporaneous rigidity may contribute to depression prevention for highly rigid individuals (Fisher et al., 2019, 2023).

This is particularly important during adolescence, a developmental period marked by profound changes across multiple domains, including in biology, interpersonal life, and stressors (school, relationships, etc.) and greater difficulties in regulating emotions (Roberts & Lopez-Duran, 2019). It is possible that high emotional rigidity reflects a specific maladaptive coping style that emerges among a subset of youth in an effort to manage this intensified emotional life (Kuppens et al., 2012). Given that adolescence is a transformative developmental period where regulation skills are also being acquired, helping adolescents develop more adaptive and flexible emotional dynamics could be crucial for long-term depression prevention. Future studies should further explore whether interventions targeting emotion rigidity contribute to improved emotion regulation and reduce the likelihood of depression onset.

This study has a few limitations. First, we only investigated six specific emotions. Including a broader range of emotions could potentially yield different density scores. Though a previous study using sixteen emotions found consistent density scores across various five-emotions network configurations (Kelley et al., 2023), we were unable to explore such network consistency with our data due to the limited number of emotions assessed. Future research incorporating a wider spectrum of emotions could not only provide a more comprehensive picture of emotional experience but also allow for separate density scores for positive and negative emotions. This would enable researchers to analyze the extent to which the association between network density and depression is specific to the interplay among negative emotions, positive emotions, or all emotions combined.

Another limitation concerns the time interval between prompts (approximately 3 h within a day). It may be that different interval lengths between two consecutive prompts would yield different results. One study found that emotion network density predicted depression and anxiety only when emotions were sampled every 90 min, not daily (Shin et al., 2022). Thus, future studies comparing different time scales could provide valuable insights into the timeframe within which emotional rigidity becomes maladaptive.

Moreover, high emotion network density is thought to reflect a self-predictive emotional system, potentially insensitive to change across situations (Bringmann et al., 2022). However, context was not explicitly modeled in this study. Individuals may experience varying degrees of situational changes during the assessment period, which could influence their emotion density scores. For instance, an individual with frequent emotional triggers might exhibit a low density score. However, this score could still be high relative to their highly fluctuating circumstances. Conversely, an individual with minimal situational changes might have a high density score that accurately reflects their emotional state. Since this study did not assess situational changes, it is important to be cautious when interpreting emotion network density as an indication of insensitivity to changing contextual demands. Future research could address this limitation by incorporating situational prompts within EMA assessments or by extending the assessment period to track

changes in emotion rigidity over longer time windows.

In addition, the mean family income in our sample was relatively high, therefore, our findings may not fully generalize to individuals from lower-income backgrounds, where differences in environmental stressors and access to resources could influence emotion dynamics and risk factors for depressive symptoms. Future research should examine whether these associations hold across more socioeconomically diverse samples. Finally, it should be noted that currently, there is no clear guideline on power analysis for a multilevel VAR model. Therefore, it is not clear what sample size and number of timepoints per subject are needed to sufficiently power these kinds of studies.

Despite these limitations, the current study extends previous research on the relationship between emotion dynamics and depression by integrating two types of emotion density (contemporaneous and temporal) as predictors of future depression risk in a sample of adolescents. The findings that emotion rigidity predicted future depression may suggest that emotion rigidity can be used to identify adolescents at high risk of developing depression and to target this maladaptive dynamic with the goal of preventing depression.

CRediT authorship contribution statement

Hadar Fisher: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Kristina Pidvirny:** Writing – review & editing, Data curation. **Nigel M. Jaffe:** Writing – review & editing, Data curation. **Anna O. Tierney:** Writing – review & editing, Data curation. **Diego A. Pizzagalli:** Writing – review & editing. **Christian A. Webb:** Writing – review & editing, Supervision, Funding acquisition.

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Declaration of competing interest

Over the past 3 years, Dr. Pizzagalli has received consulting fees from Boehringer Ingelheim, Compass Pathways, Engrail Therapeutics, Karla Therapeutics, Neumora Therapeutics (formerly BlackThorn Therapeutics), Neurocrine Biosciences, Neuroscience Software, Otsuka, Sage Therapeutics, Sama Therapeutics Sunovion, and Takeda; he has received honoraria from the Psychonomic Society, Springer, and American Psychological Association (for editorial work) and from Alkermes; he has received research funding from the Bird Foundation, Brain and Behavior Research Foundation, Dana Foundation, Millennium Pharmaceuticals, NIMH, and Wellcome Leap; he has received stock options from Compass Pathways, Engrail Therapeutics, Neumora Therapeutics, and Neuroscience Software. Dr. Webb receives consulting fees from King & Spalding law firm. No funding from these entities was used to support the current work, and all views expressed are solely those of the authors. All other authors have no conflicts of interest or relevant disclosures.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.brat.2025.104818.

Data availability

Data will be made available on request.

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